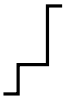
Math Kangaroo Practice

Grade 3-4 · 24 questions · harder · seed 57313

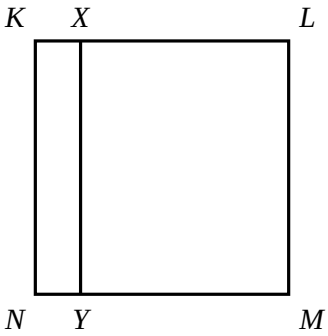
Part A: Each question is worth 3 points

1. Far away we see the skyline of a castle.

Which of the pieces **cannot** belong to the skyline?

A. ***B.*** ***C.*** ***D.***  ***E.***

2. *KLMN* is a square. Its side is 28 cm. The line *XY* cuts it into two rectangles, and *KX* = 5 cm.

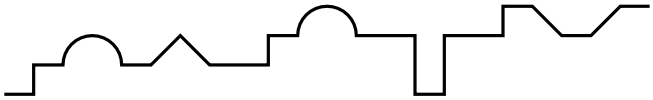


How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *XLMY*?

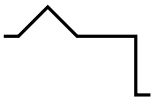
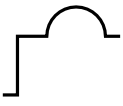
A. 23 cm ***B.*** 10 cm ***C.*** 9 cm ***D.*** 8 cm ***E.*** 5 cm

Part B: Each question is worth 4 points

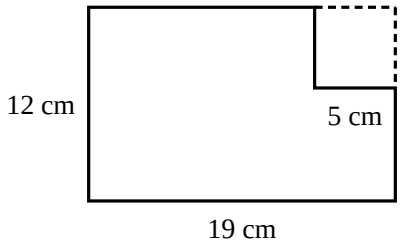
3. This is the outline of a town seen from the river at sunset.



Which piece **can** be part of the outline?

A. ***B.***  ***C.*** ***D.*** ***E.*** 

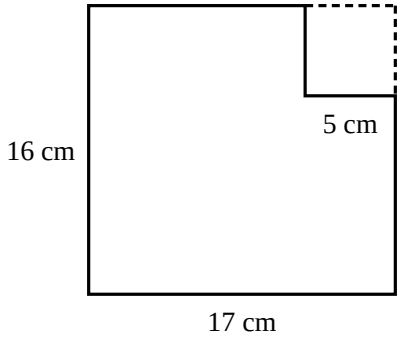
4. A rectangle is 19 cm long and 12 cm wide. A square with side 5 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 82 cm ***B.*** 72 cm ***C.*** 62 cm ***D.*** 57 cm ***E.*** 52 cm

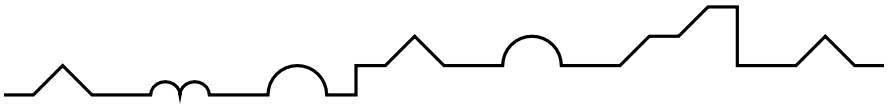
5. A rectangle is 17 cm long and 16 cm wide. A square with side 5 cm is cut off one corner.



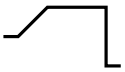
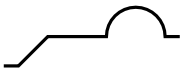
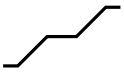
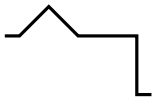
What is the perimeter of the remaining shape?

A. 66 cm ***B.*** 61 cm ***C.*** 56 cm ***D.*** 46 cm ***E.*** 33 cm

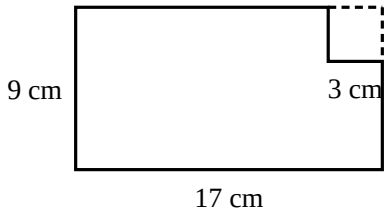
6. Far away we see the skyline of a castle.



Which of the pieces **can** belong to the skyline?

A.  ***B.***  ***C.***  ***D.*** ***E.*** 

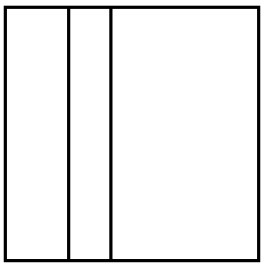
7. A rectangle is 17 cm long and 9 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 64 cm ***B.*** 52 cm ***C.*** 49 cm ***D.*** 46 cm ***E.*** 40 cm

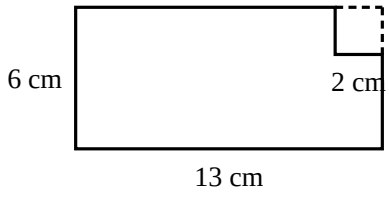
8. A square has perimeter 52 cm. It is cut along straight lines into 3 rectangles as shown.



What is the sum of the perimeters of all 3 rectangles?

A. 156 cm ***B.*** 118 cm ***C.*** 104 cm ***D.*** 91 cm ***E.*** 78 cm

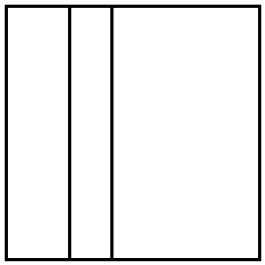
9. A rectangle is 13 cm long and 6 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 47 cm ***B.*** 46 cm ***C.*** 42 cm ***D.*** 39 cm ***E.*** 38 cm

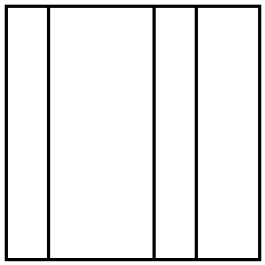
10. A square has perimeter 32 cm. It is cut along straight lines into 3 rectangles as shown.



What is the sum of the perimeters of all 3 rectangles?

A. 96 cm ***B.*** 67 cm ***C.*** 64 cm ***D.*** 56 cm ***E.*** 48 cm

11. A square garden is surrounded by a fence 28 m long. The gardener divides it into 4 rectangular beds as shown and puts a separate fence around each bed.

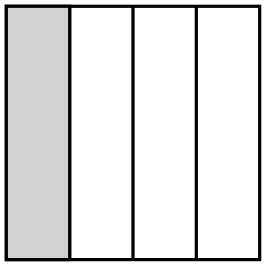


How many metres of fence are around the beds altogether?

A. 42 m ***B.*** 49 m ***C.*** 70 m ***D.*** 86 m ***E.*** 112 m

Part C: Each question is worth 5 points

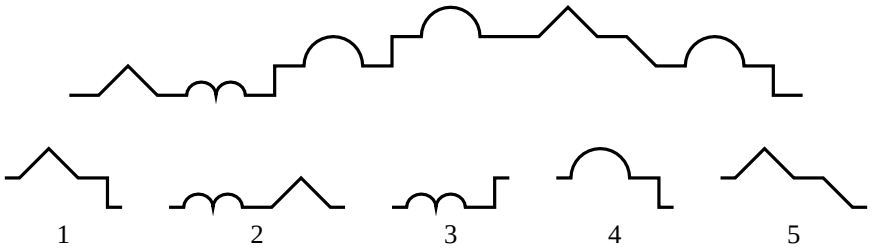
12. Mia cuts a square sheet of paper into 4 equal strips as shown. The perimeter of each strip is 60 cm.



How long is the side of the square?

A. 96 cm ***B.*** 36 cm ***C.*** 30 cm ***D.*** 24 cm ***E.*** 6 cm

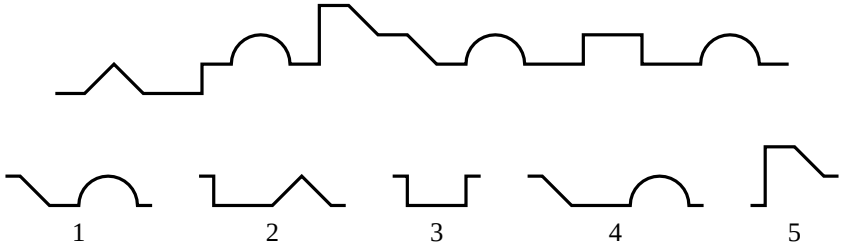
13. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

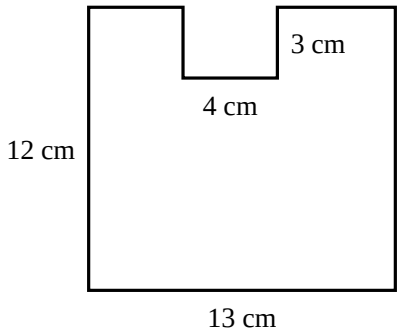
14. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

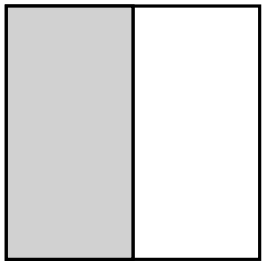
15. A 13 cm by 12 cm rectangle of cardboard has a rectangular notch 4 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 53 cm ***B.*** 56 cm ***C.*** 58 cm ***D.*** 64 cm ***E.*** 67 cm

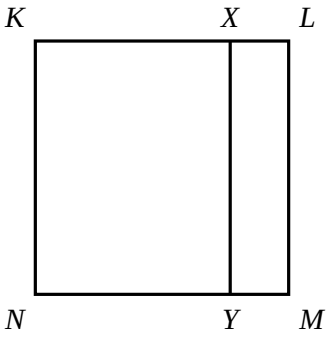
16. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 36 cm.



What is the perimeter of the square?

A. 72 cm ***B.*** 48 cm ***C.*** 44 cm ***D.*** 24 cm ***E.*** 12 cm

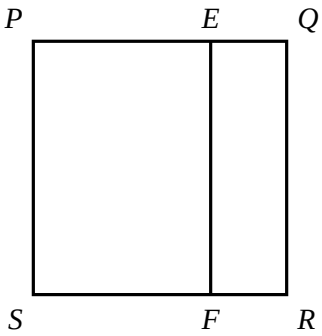
17. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 46 cm and the perimeter of *XLMY* is 32 cm.



What is the perimeter of the square *KLMN*?

A. 65 cm ***B.*** 52 cm ***C.*** 39 cm ***D.*** 19 cm ***E.*** 14 cm

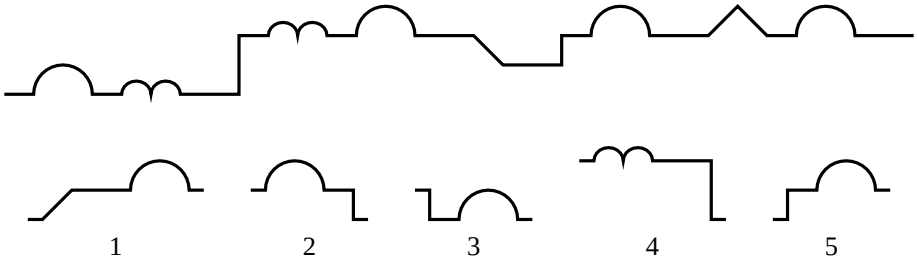
18. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 34 cm and the perimeter of *EQRF* is 26 cm.



How long is the side of the square *PQRS*?

A. 6 cm ***B.*** 7 cm ***C.*** 8 cm ***D.*** 9 cm ***E.*** 10 cm

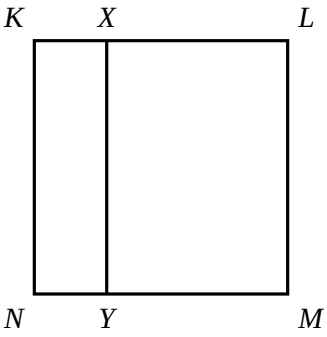
19. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

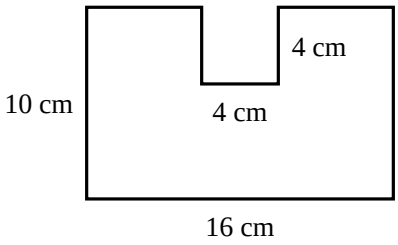
20. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 18 cm and the perimeter of *XLMY* is 24 cm.



What is the perimeter of the square *KLMN*?

A. 7 cm ***B.*** 14 cm ***C.*** 21 cm ***D.*** 28 cm ***E.*** 42 cm

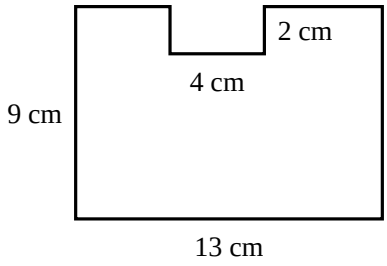
21. A 16 cm by 10 cm rectangle of cardboard has a rectangular notch 4 cm wide and 4 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 48 cm ***B.*** 52 cm ***C.*** 60 cm ***D.*** 61 cm ***E.*** 68 cm

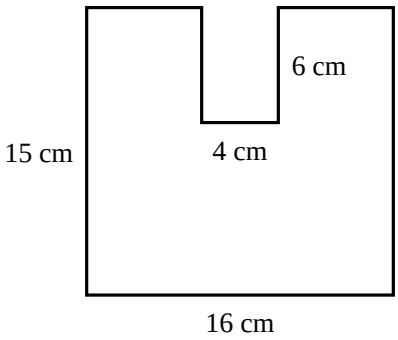
22. A 13 cm by 9 cm rectangle of cardboard has a rectangular notch 4 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 44 cm ***B.*** 48 cm ***C.*** 52 cm ***D.*** 53 cm ***E.*** 56 cm

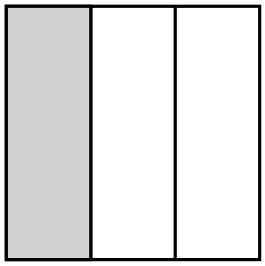
23. A 16 cm by 15 cm rectangle of cardboard has a rectangular notch 4 cm wide and 6 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 58 cm ***B.*** 62 cm ***C.*** 68 cm ***D.*** 70 cm ***E.*** 74 cm

24. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 64 cm.



How long is the side of the square?

A. 8 cm ***B.*** 16 cm ***C.*** 18 cm ***D.*** 20 cm ***E.*** 24 cm

Answer Key

1.	**E**	**2.**	**B**	**3.**	**A**	**4.**	**C**	**5.**	**A**	**6.**	**C**
7.	**B**	**8.**	**C**	**9.**	**E**	**10.**	**C**	**11.**	**C**	**12.**	**D**
13.	**C**	**14.**	**C**	**15.**	**B**	**16.**	**B**	**17.**	**B**	**18.**	**E**
19.	**A**	**20.**	**D**	**21.**	**C**	**22.**	**B**	**23.**	**E**	**24.**	**E**

Solutions

1. E — All the other pieces can be found in the line, but this one cannot: it has a round dome where the line has a pointed roof. [2002 Q7 · cannot_belong]

2. B (10 cm) — *XLMY* has sides 23 and 28 cm, perimeter 102 cm. Square: 112 cm. Difference 10 cm (= 2 × 5). [2002 Q6 · square_minus_strip]

3. A — Only this piece is found in the line. The others don't fit — B: it has a pointed roof where the line has a round dome; C: it has the right shapes but in the wrong order; D: it is a mirror image of a part of the line; E: its step is taller than in the line. [2002 Q7 · can_belong]

4. C (62 cm) — The two new edges (5 cm and 5 cm) replace the two removed pieces of the outer edge, which were also 5 cm and 5 cm. So the perimeter is the same as the rectangle's: 2 × (19 + 12) = 62 cm. [2002 Q6 · corner_cut]

5. A (66 cm) — The two new edges (5 cm and 5 cm) replace the two removed pieces of the outer edge, which were also 5 cm and 5 cm. So the perimeter is the same as the rectangle's: 2 × (17 + 16) = 66 cm. [2002 Q6 · corner_cut]

6. C — Only this piece is found in the line. The others don't fit — A: its flat part between the two features has the wrong length; B: it has the right shapes but in the wrong order; D: its flat part between the two features has the wrong length; E: it has the right shapes but in the wrong order. [2002 Q7 · can_belong]

7. B (52 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (17 + 9) = 52 cm. [2002 Q6 · corner_cut]

8. C (104 cm) — Side = 52 ÷ 4 = 13 cm. There are 2 cuts; each cut is 13 cm long and is counted twice (once for each piece next to it): 52 + 2 × 2 × 13 = 104 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

9. E (38 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (13 + 6) = 38 cm. [2002 Q6 · corner_cut]

10. C (64 cm) — Side = 32 ÷ 4 = 8 cm. There are 2 cuts; each cut is 8 cm long and is counted twice (once for each piece next to it): 32 + 2 × 2 × 8 = 64 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

11. C (70 m) — Side = 28 ÷ 4 = 7 m. There are 3 cuts; each cut is 7 m long and is counted twice (once for each piece next to it): 28 + 3 × 2 × 7 = 70 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

12. D (24 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 60 ÷ 10 = 6, so the side is 4 × 6 = 24 cm. [2002 Q6 · strips_to_square]

13. C (3) — Pieces 3, 4, 5 fit. Not: 1 (it has a pointed roof where the line has a round dome); 2 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

14. C (2) — Pieces 1, 5 fit. Not: 2 (it has a pointed roof where the line has a round dome); 3 (it has the right shapes but in the wrong order); 4 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

15. B (56 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (13 + 12) + 2 × 3 = 56 cm. [2002 Q6 · notch]

16. B (48 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 36 ÷ 6 = 6, so the side is 2 × 6 = 12 cm and the perimeter is 48 cm. [2002 Q6 · strips_to_square]

17. B (52 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 46 + 32 = 78 cm = 6 sides. Side = 78 ÷ 6 = 13 cm, perimeter 4 × 13 = 52 cm. [2002 Q6 · two_perimeters]

18. E (10 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 34 + 26 = 60 cm = 6 sides. Side = 60 ÷ 6 = 10 cm. [2002 Q6 · two_perimeters]

19. A (1) — Only piece 5 fits. Not: 1 (it is a mirror image of a part of the line); 2 (it is a mirror image of a part of the line); 3 (it steps down where the line steps up); 4 (it steps down where the line steps up). [2002 Q7 · how_many_fit]

20. D (28 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 18 + 24 = 42 cm = 6 sides. Side = 42 ÷ 6 = 7 cm, perimeter 4 × 7 = 28 cm. [2002 Q6 · two_perimeters]

21. C (60 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (16 + 10) + 2 × 4 = 60 cm. [2002 Q6 · notch]

22. B (48 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (13 + 9) + 2 × 2 = 48 cm. [2002 Q6 · notch]

23. E (74 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 6 cm. Perimeter = 2 × (16 + 15) + 2 × 6 = 74 cm. [2002 Q6 · notch]

24. E (24 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 64 ÷ 8 = 8, so the side is 3 × 8 = 24 cm. [2002 Q6 · strips_to_square]